



Introduction to Bioengineering Theory (BBL1020; 3-0-0)

Lecture-1: 3 Classes

Google Classroom code: k4m2ay6p

*Please listen carefully
&
Keep your phone(s) on silent mode*

Objectives

1. Provide a strong foundation of biology in the context of engineering by highlighting key differences in the approaches of both disciplines, while illustrating how bioengineering utilizes the best of both disciplines.
2. Provide a multidisciplinary learning.

Learning Outcomes

The students are expected to have the ability to:

1. Approach biology from an engineering perspective by gaining an understanding of how biological systems function.
2. Be inspired from biological systems and processes that have been perfected by evolution.
3. Conceptualize novel engineering solutions for complex biological problems.

Course Syllabus



Biology & Engineering Approaches [3 lectures]: Observations; deductive, inductive and abductive reasoning; hypothesis testing; reductionism; systems biology; modeling; tinkering with biological systems; cause & effect; translational research; attributes of a bioengineer.

Origin & Evolution of Life [7 lectures]: Chemical basis of life, water sustains life, biomolecules, biological macromolecules, unity in diversity among life forms, phylogeny.

Functioning of Biological Systems [8 lectures]: From organisms to cells, structure-function relationships, free energy, entropy, reactions, biocatalysis, energy coupling, effects of process failure.

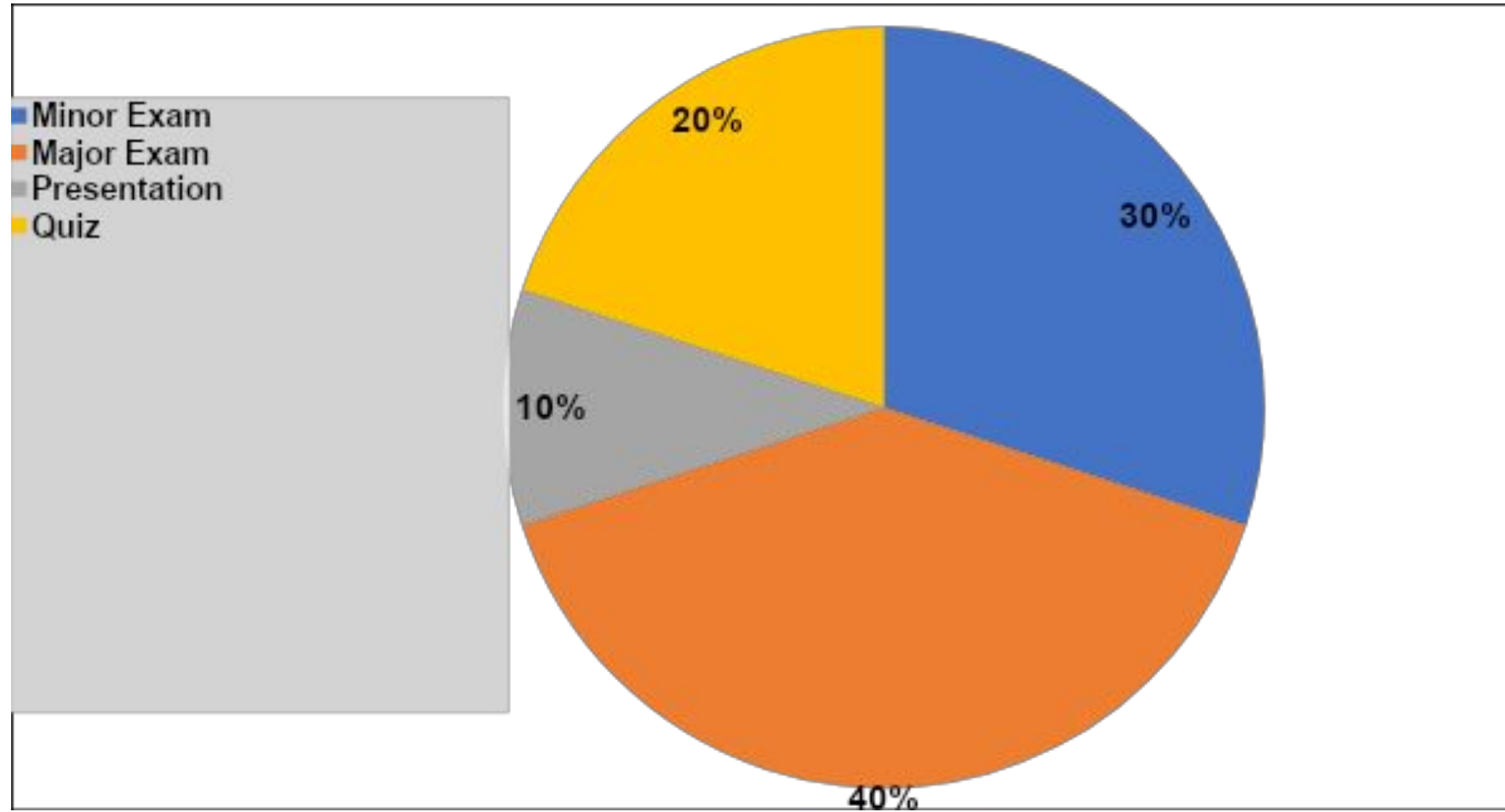
Probability in life processes [7 lectures]: Storage of genetic information, central dogma of biology, cell division, inheritance, genetic variation.

Control in living systems [6 lectures]: Gene expression logic gates, control of cell division, hormones, signals & receptors, feedback control, control failure & disease.

Responses of Biological systems [4 lectures]: Interactions, communication, stress responses, tolerance, adaptation, rhythms & cycles, ecosystems.

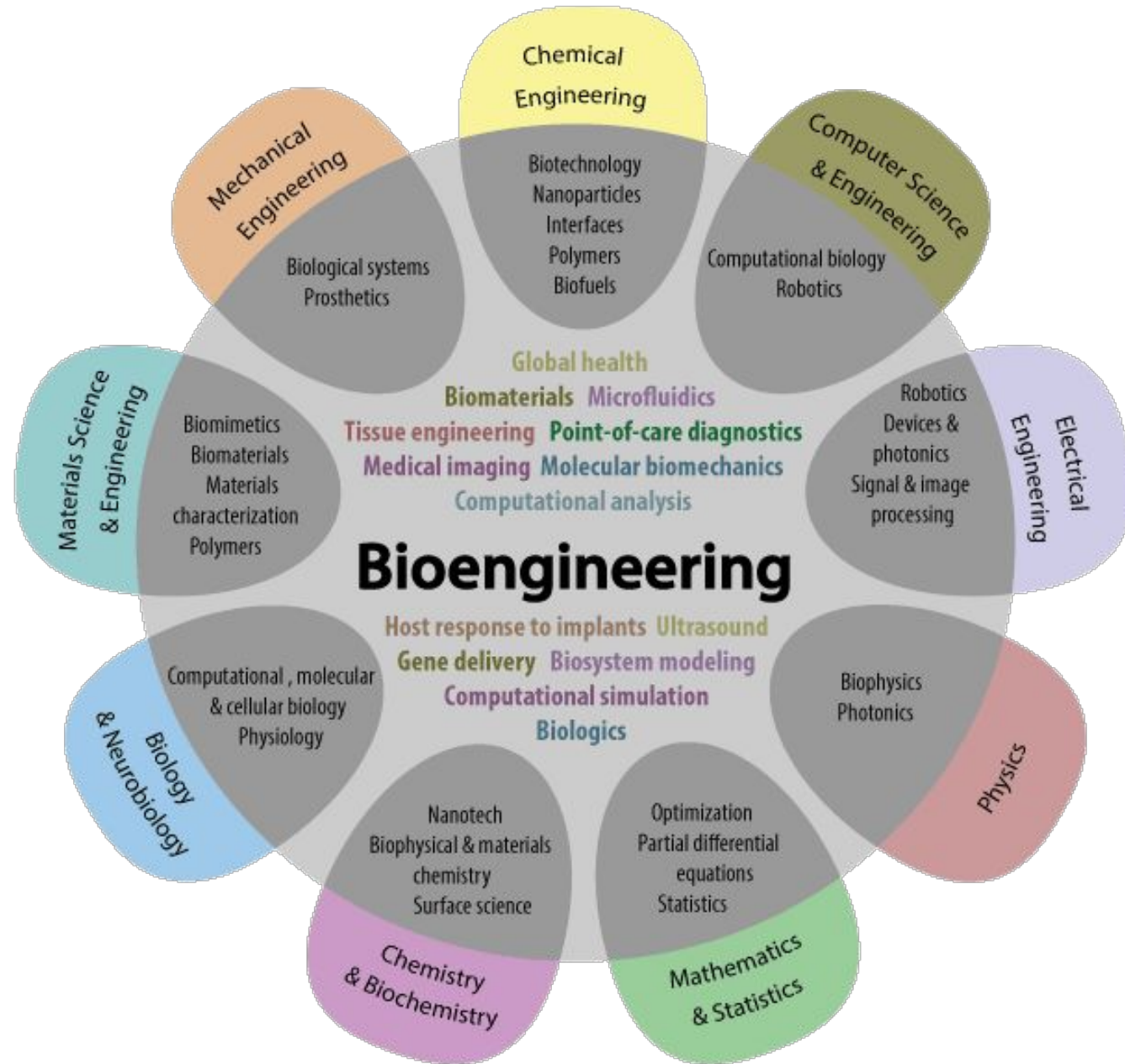
Exploitation of living systems [5 lectures]: Antibiotics, Recombinant DNA Technology, Biomaterials-bioplastics and materials for healthcare, Biodiesel and drop in biofuels from algae.

Evaluation Chart



Biotechnology	Biotechnology is a multidisciplinary field that utilizes biological systems, living organisms, or parts thereof (e.g., cells, enzymes, DNA) to develop or create products and technologies for specific purposes. It integrates natural sciences (like molecular biology, genetics, microbiology) and engineering sciences.
Bioengineering	Bioengineering is the application of engineering principles and design concepts to biological systems. It aims to understand, modify, or control living systems and to design and develop products or processes that interact with biological systems. It's a broad umbrella term that applies engineering methodologies to any biological problem.
Biomedical Engineering	Biomedical engineering (BME) is the application of engineering principles and design concepts to medicine and biology for healthcare purposes. It is a subset of bioengineering.
Biological Engineering	Largely synonymous with "Bioengineering." It's the application of engineering principles and tools to solve problems across all biological systems. It leverages knowledge from physical sciences, mathematics, and biology to create usable, tangible, and economically viable products and solutions. Emphasizes the fundamental engineering principles applied to biological systems at various scales, from molecular to ecosystem levels.
Bioinspired Engineering	Bioinspired engineering is an approach to innovation that seeks sustainable solutions to human challenges by emulating nature's time-tested patterns and strategies. Instead of applying engineering to biology, it applies lessons from biology to engineering design. Nature serves as the blueprint or mentor for solving engineering problems.

000



A story to motivate to be here

000

-story of undaunted spirit and extraordinary courage. Despite losing their hands or legs, they did not give up.
- There are millions like them in this country across the globe.

How can Science and Technology help them to live a better life?



Conventional prosthetic limbs: Replace the form and function of the absent limb

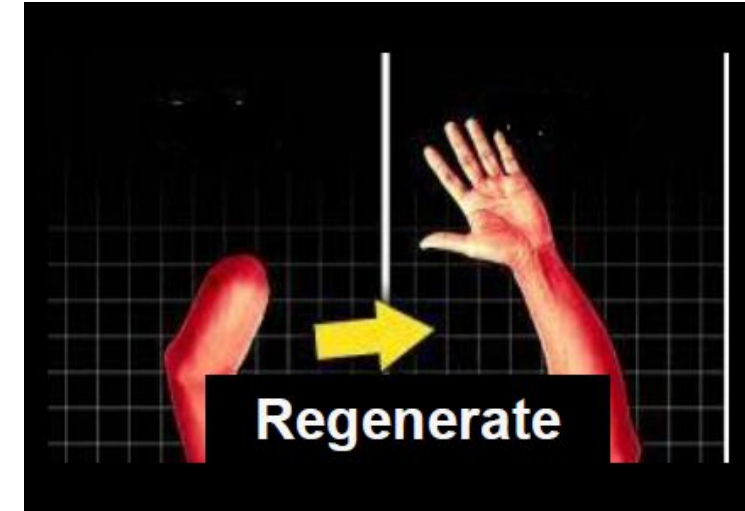


Bionic limbs: Controlled by individual's biosignals and enable seamless movement.

<https://synergypo.com/blog/main-types-of-artificial-limbs-prosthesis/>

What if we can regenerate an amputated limb?

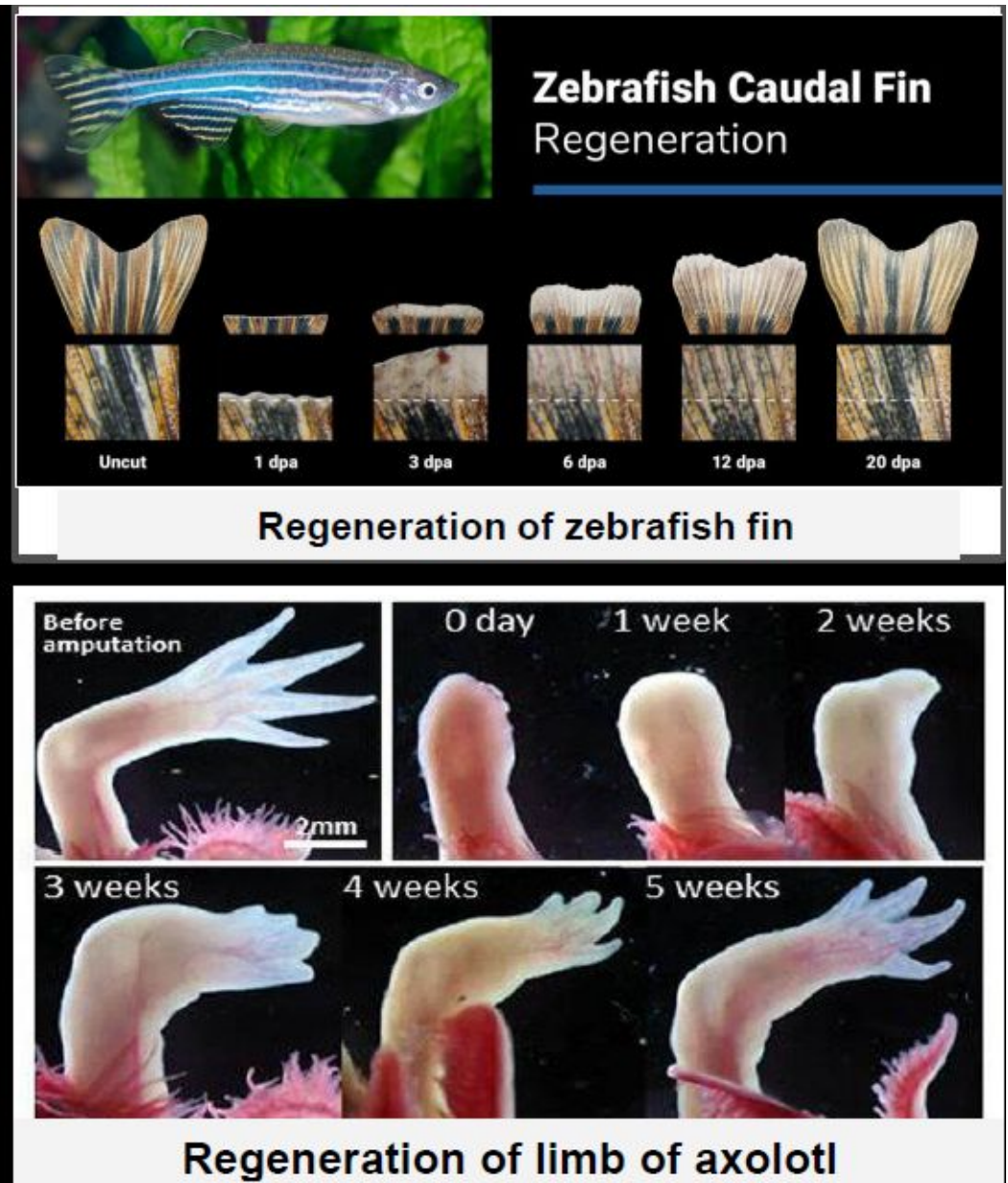
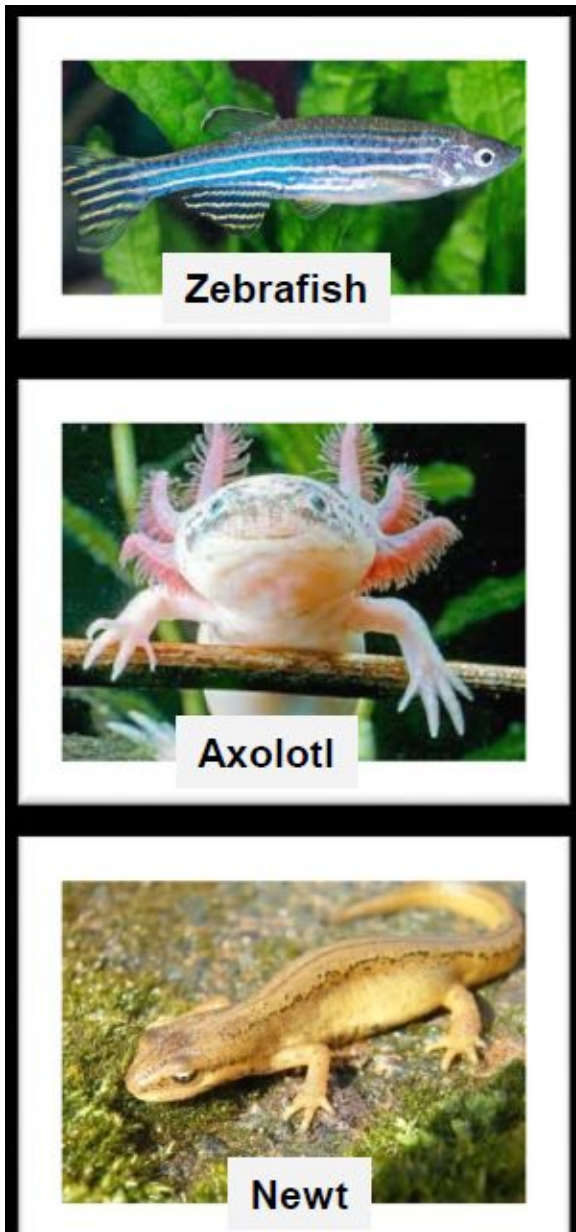
000



Looks like a science fiction for now. But can it be possible in future??



Some animals
possess remarkable
capacity of
regeneration



Understand Nature to create Engineering Solutions

**“Nature has no departments of chemistry, biology, physics, or arts”.
The only thing is known for Nature is **“ADAPTATION”**”**

Tardigrades



Cart Wheeling Spider



Marine mussel



Fruit fly



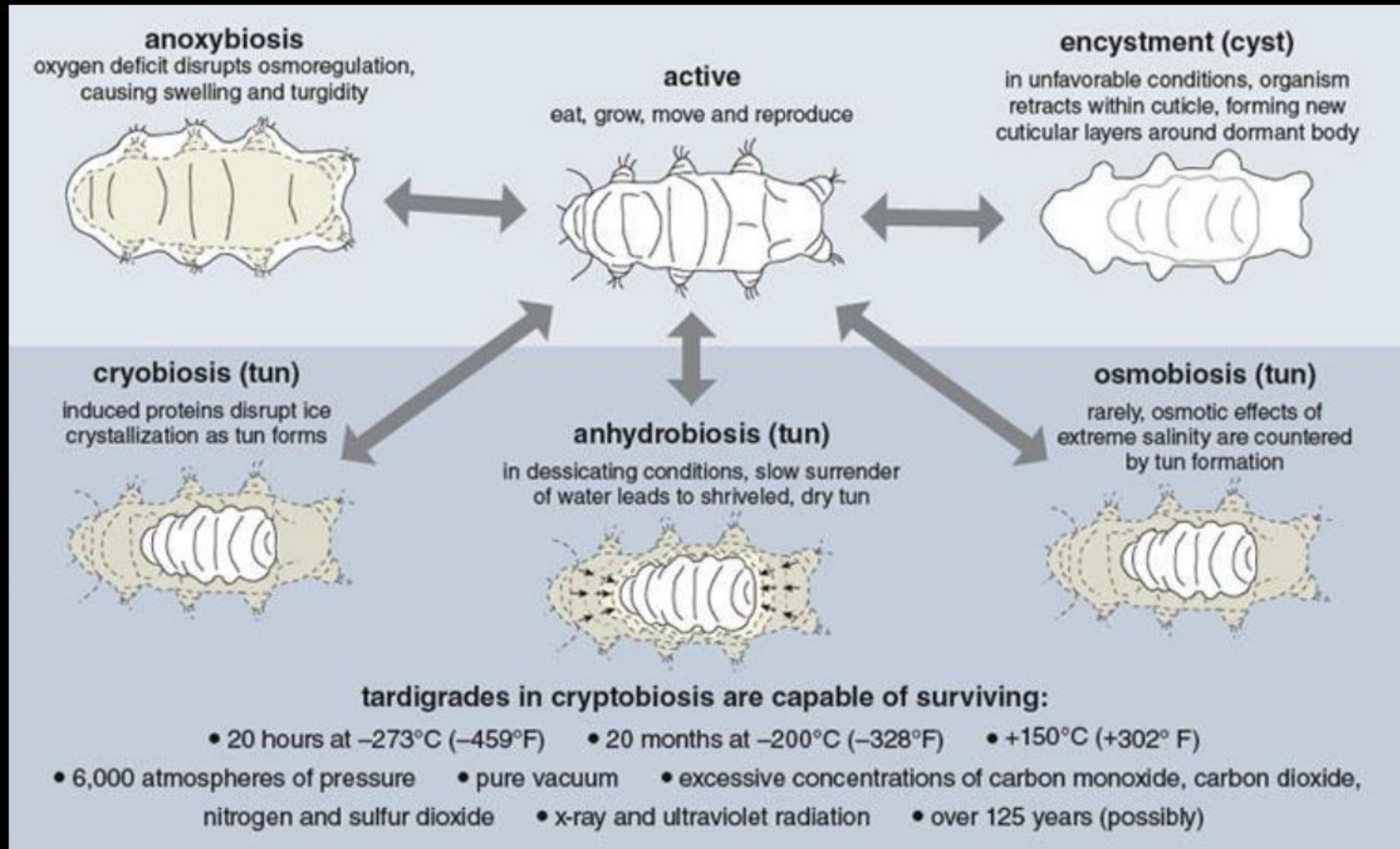
Bombardier Beetle



Biological solutions and limitless possibilities



Tardigrades



Cartwheeling spiders

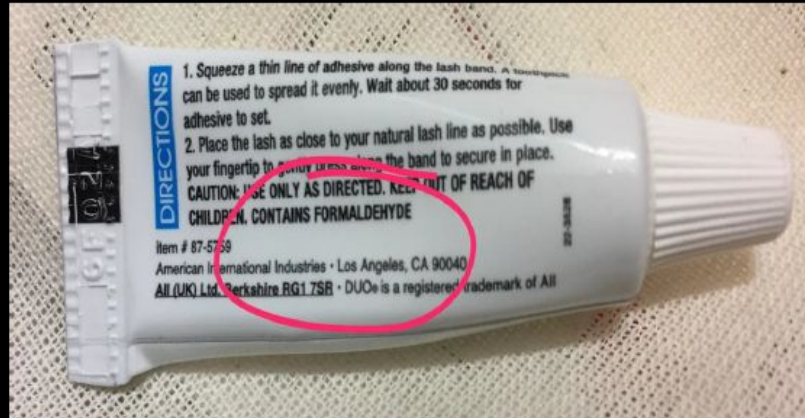


Cartwheeling spiders, like the [Golden Wheel Spider](#) and the [Flick-Flack Spider](#) have evolved unique adaptations for desert survival, primarily **cartwheeling** as a rapid escape from predators (especially [pompilid wasps](#)) by tucking into a ball and rolling down dunes, achieving high speeds. Other adaptations include **silk-lined burrows for cooling**, a **trapdoor entrance**, and even using their hind legs to launch at threats or create silk traps.

Sticky business with mussel



Commercial glue



Contains formaldehyde:
carcinogenic

Ineffective under water

Marine mussel

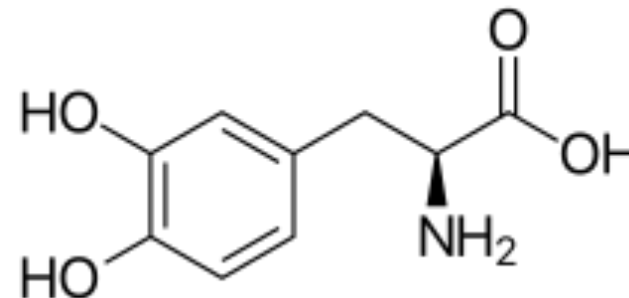


Underwater adhesion:

adhesive protein that contains
a unique amino acid (DOPA)
with a catechol side chain
responsible for strong
adhesion to underwater
surface

Smart underwater adhesive!

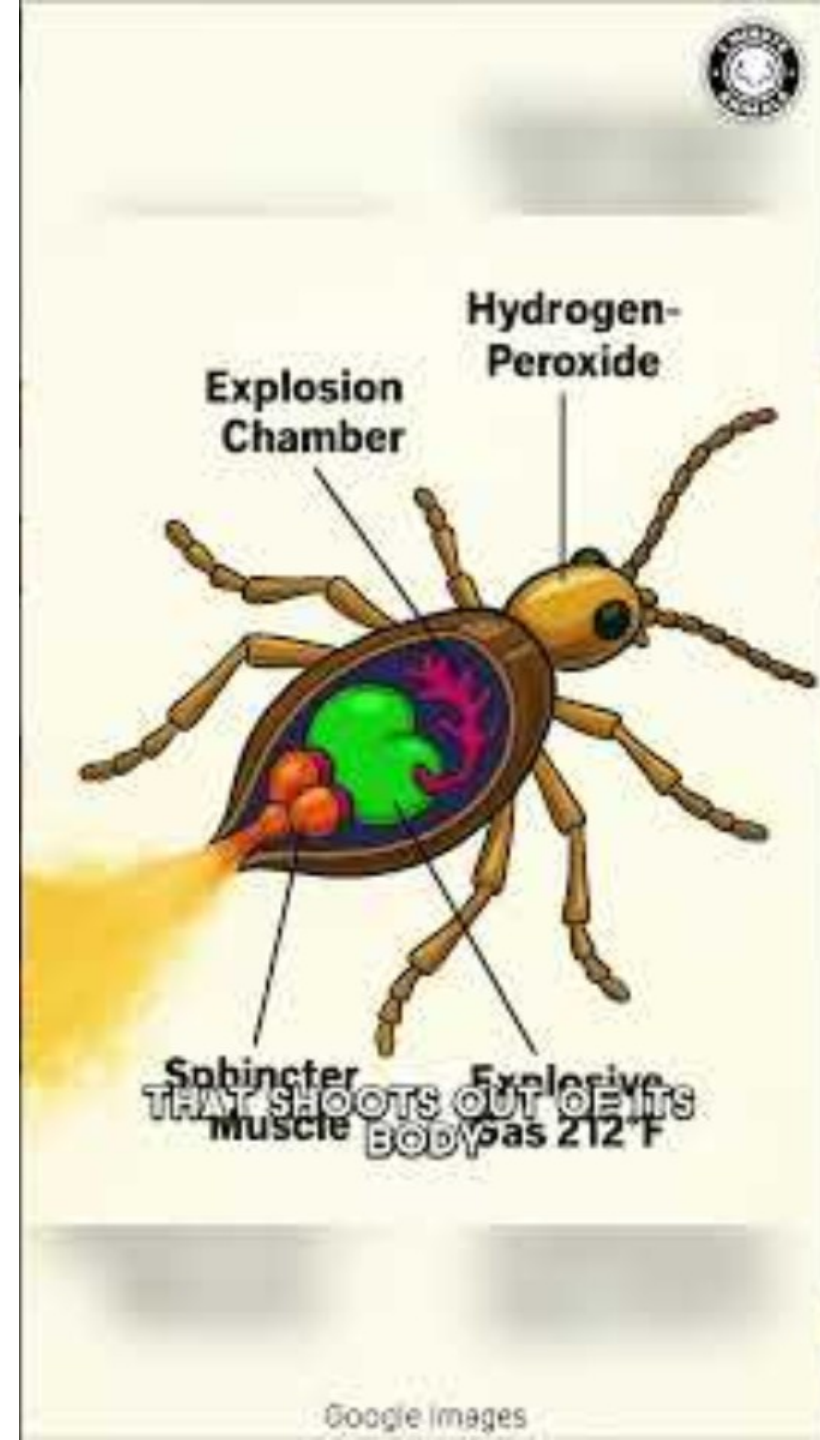
L-3,4-dihydroxyphenylalanine (L-DOPA)



Bombardier beetles

Bombardier beetles have a unique adaptation: they spray predators with a boiling, noxious chemical mixture from their abdomen, produced by an explosive reaction between hydroquinone and hydrogen peroxide, catalyzed by enzymes, creating scalding liquid and gas.

Key features include specialized glands, controlled bursts (up to 500/second), a rotating nozzle for aiming, and a pulsing mechanism to cool the system and prevent self-injury, making it a highly sophisticated chemical defense



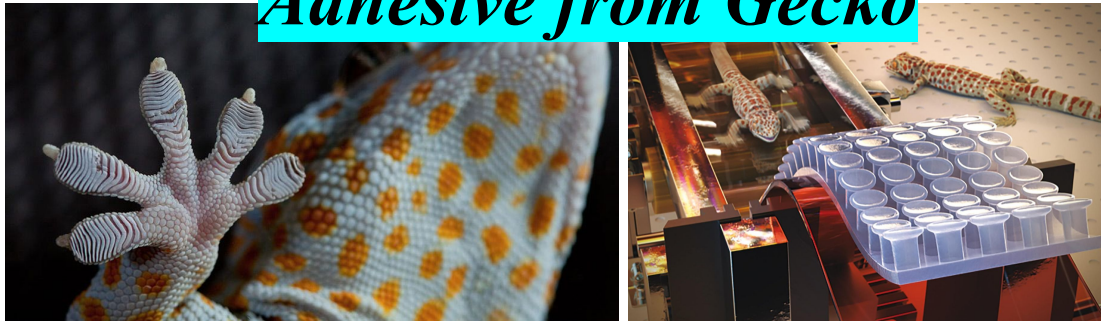
Nature knows the secret and it inspire solutions



Velcro from Burr



Adhesive from Gecko



The Shinkansen Bullet Train inspired by Kingfisher bird



<http://creativecommons.org/2.0/License>



Turbine from Whale

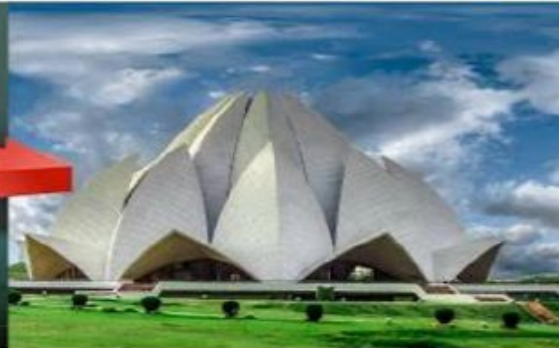
Box fish to Nextgen Merc- Bionic



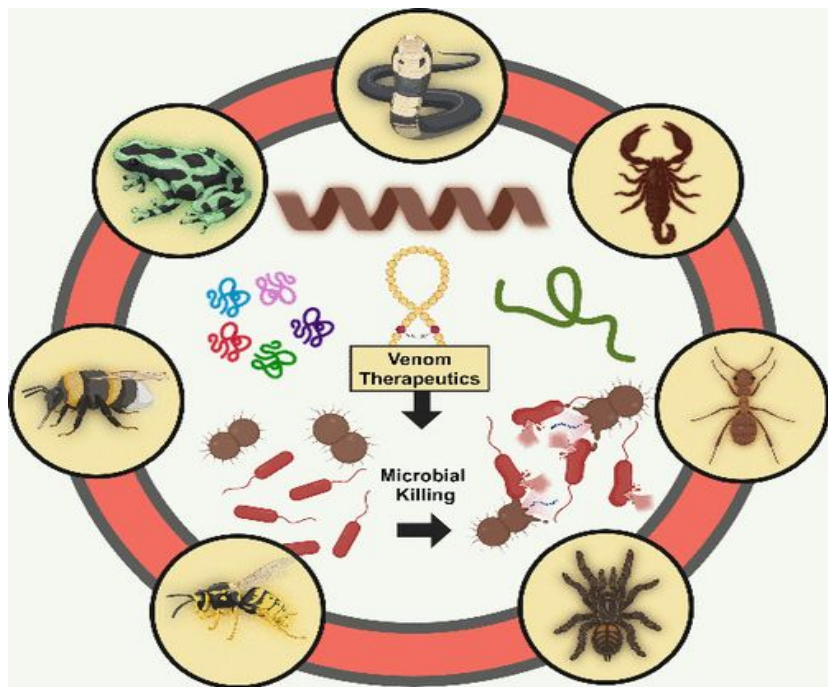
Termite houses to AC free buildings



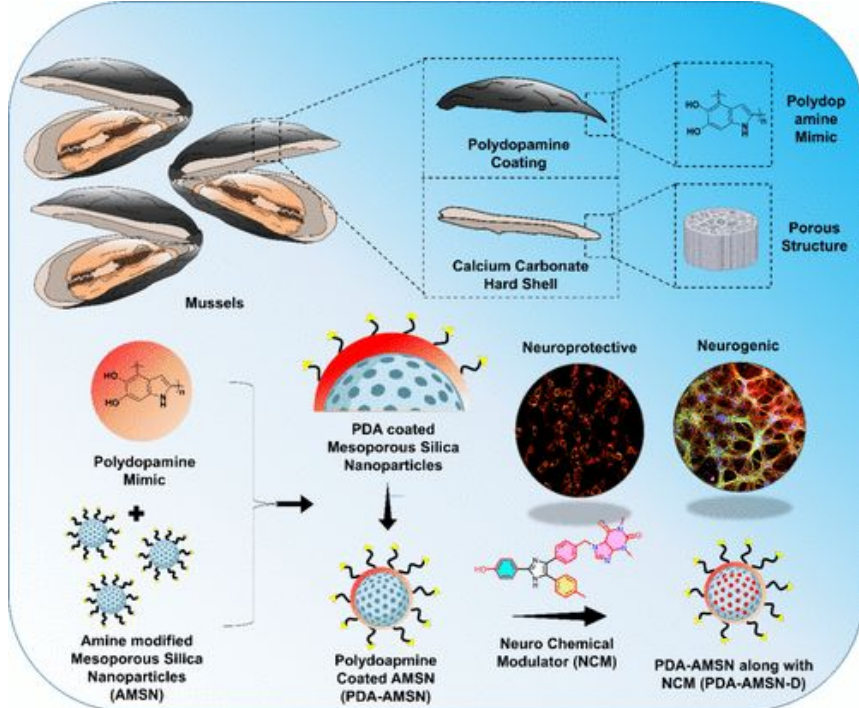
Biomimicry in Architecture



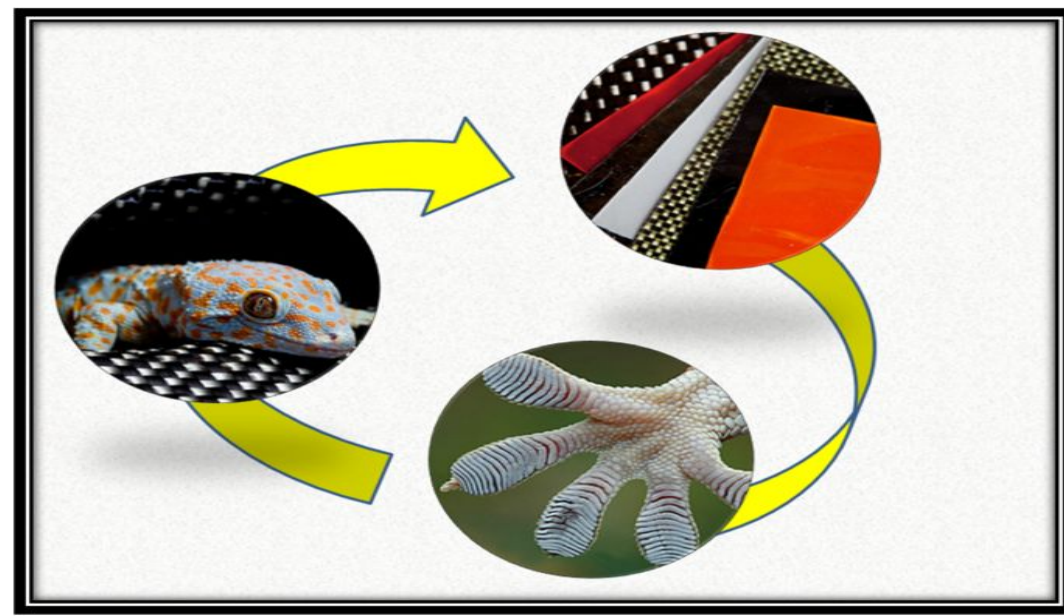
Antimicrobial Therapeutics



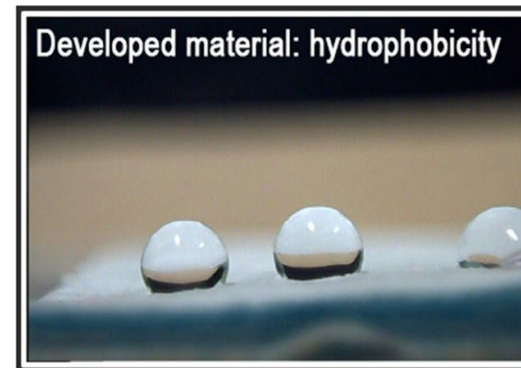
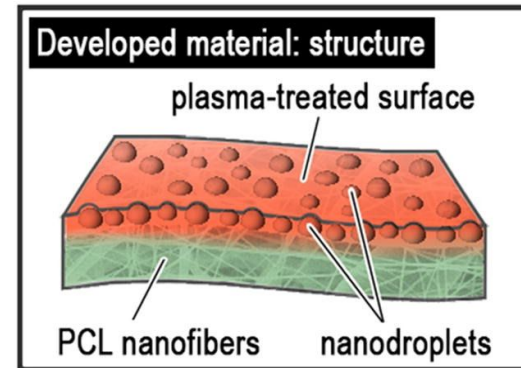
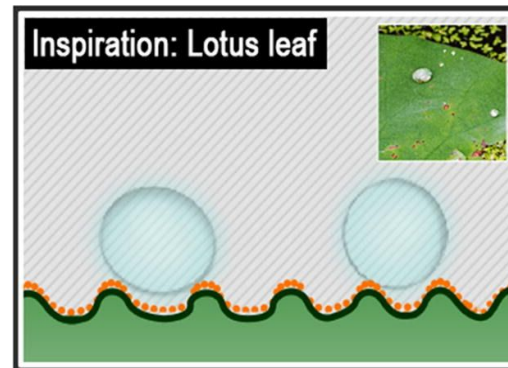
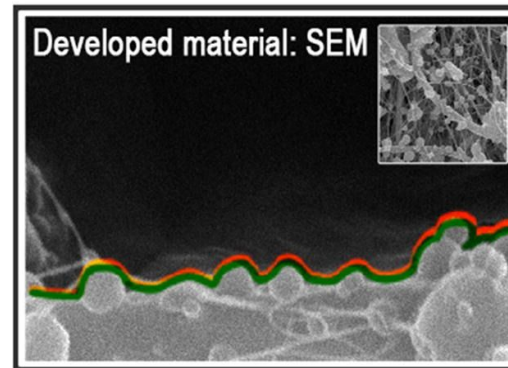
Mussel-Inspired Polydopamine Nanocomposite



bandages inspired by Gecko's feet



Hydrophobic materials inspired by Lotus leaf



Visit: asknature.org

ask nature

COLLECTIONS

BIOLOGICAL STRATEGIES

INNOVATIONS

FOR EDUCATORS

ABOUT

SEARCH

Biological Strategies

+

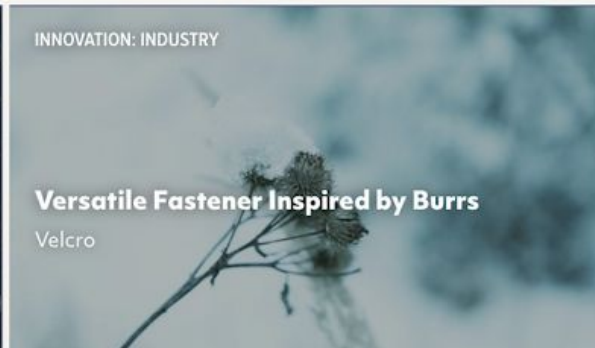
[Explore Biological Strategies](#)



Innovations

+

[Explore Innovations](#)



ANY

QUESTIONS?